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Objective and Topics : Week 18 apply responsible AI thinking and selected the below 2 use cases a real world failure and Part 2 I have used my organizational Chatbot example. For Privacy and ethical considerations, I have given limited information for Part 2 as I need to abide by the internal policies.

Part 1: Real-World AI Failure (300–400 words)

- **Case Selected:** [Name of system + year + where used]
- **Context:** [What the system was designed to do; stakeholders; decision setting]
- **Failure Description:** [What went wrong; how it showed up; what “failure” means here]
- **Impact:** [Who was affected; type of harm; ethical implications]
- **Root Cause:** [Technical + organisational factors; why safeguards failed or were missing]
- **Preventive Actions:** [3–6 specific mitigations: testing/audits/oversight/monitoring/governance]
- **Sources:** [Links or citations]

Case Selected:

Amazon developed an internal AI-powered recruiting tool to automate resume screening. However, the system learned gender bias from historical hiring data and systematically penalized women candidates, leading to its eventual shutdown.

Context : Internal hiring system at Amazon (2014–2017). Resume screening for technical roles (e.g., software engineers). Ranking candidates using a 1–5 star scoring system

Failure Description: Bias and penalized the resume which contain women related applications. machine-learning specialists uncovered a big problem: their new recruiting engine did not like women.

Specifics over slogans: The system did things wrong as it Penalized resumes containing: Words like “women’s” (e.g., women’s chess club) . Favored resumes resembling male-dominated historical profiles. Failed to evaluate candidates gender-neutrally. Reinforced existing hiring bias instead of removing it

Mechanism and root cause: Model trained on 10 years of historical resumes. Majority of resumes came from male candidates ,No fairness constraints or bias correction mechanisms. Optimization focused only on pattern matching, not fairness

Impact and ethics: Explain ability tools, Audit pipelines during development, Data reflects male dominance in tech hiring. Penalizes resumes with female indicators. Amazon abandons the system entirely. Real impact was on Women candidates as they were Unfairly downgraded or filtered out, Loss of Equal opportunity, lack of Trust in hiring systems. Reputational risk: Highlighted globally as AI bias example, Slowed adoption of AI in hiring. Project scrapped after years of investment. Ethical Impact : Trust and Transparency was missing. AI systems perceived as objective, but were biased. Violated equal employment principles

Preventive actions:

Removing gender-indicative terms. Reweighting features, Amazon later introduced tools like bias detection in ML pipelines (e.g., SageMaker Clarify).

Data audits before training, Model fairness validation, Responsible AI frameworks and Ethical review boards

Continuous monitoring for Drift , and then LLM-as-judge / fairness evaluators (modern approach).

Sources:

https://mediawell.ssrc.org/news-items/amazon-scrap-secret-ai-recruiting-tool-that-showed-bias-against-women-reuters/?utm_source=chatgpt.com

<https://www.reuters.com/article/us-amazon-com-jobs-automation-insight/amazon-scrap-secret-ai-recruiting-tool-that-showed-bias-against-women-idUSKCN1MK08G/>

Part 2: Safeguards for an Agent in Your Domain (300–400 words)

- **Domain and Use Case:** [Who uses it; what it does; what it should not do] – Domain is AgriTech and Use Case is internal IT ChatBOT which processes information related to the knowledge base within the organization.
- **Key Risk Areas:** [Top 2–3 risks specific to this use case]
 - Risk #1 : Data exposure and confidentiality breach : The most immediate risk is the employees using or sharing any confidential , personal , proprietary data with the AI Tools. Data feeding to chatgpt or external tools is strictly prohibited which sit outside the organizational domain or boundaries. Monitoring of the customer sensitive data is of importance and guardrails are in place , also it is important to

do so as AI tools can leak data which can have legal, competitive and regulatory consequences

- Risk #2 : Overreliance on the AI Tools Output(Hallucination Risk) : AI models can generate responses that are plausible but factually incorrect which is know as hallucination. In business context this is dangerous in certain business areas for process guidance, regulatory compliance, master data decisions, planning where this data needs to be validated and verified which is given as AI output. Outdated and incomplete data can give misleading outputs.
- **Safeguard 1 – Data Privacy:** [What data is sensitive; how you protect it; how consent/access works]
 - Data Privacy aspects for the internal platforms are build on the Microsoft Azure infrastructure which means the data stays within the organizations enterprise boundary and does or is not used to train the external models. Employees are expected to follow the existing data classification standards i.e internal, and public data must be handled very carefully before they are injected for any AI tools.
 - Personal Identifiable information (PII) are governed by the EU GDPR standards and this data or business sensitive data should not be entered / injected into the tool before proper approvals are taken internally.
- **Safeguard 2 – Content Safety:** [What unsafe outputs look like; how you prevent/limit them; how you handle uncertainty]
 - For the AI tools which are deployed they undergo the validation and verification checks and governance process has to be followed by the development or deployment teams
 - For knowledge base tools content has to be managed through a internal data life cycle process and archival is of utmost importance
 - Designated owners and review schedules to be followed
 - Data content filters provided by Microsoft Azure Open AI should also apply for blocking harmful, offensive or any policy violating outputs
- **Safeguard 3 – Operational Oversight:** [Human review points; audit logs; monitoring; escalation and rollback]
 - AI tool deployment , usage and quality metrics should be tracked oftenly.
 - Knowledge management framework provides a framework to have the AI systems consume reliable, structured and governed content.
 - Human oversight is maintained for making informed critical decisions e.g in case of Incident data.
 - Audit logs and monitoring is governed by the framework and policies